

Practice problems-1

Problem 1

A wave travelling along a string is given by

$$y(x,t) = 2 \sin(4\pi t + 10\pi x) \text{ (cm)},$$

where x is the distance along the string in metres and y is the vertical displacement.

Determine

- the direction of wave travel
- the reference phase ϕ_0
- the frequency
- the wavelength and
- phase velocity

a) Writing the eq in terms of $\cos$

$$y(x,t) = 2 \cos(4\pi t + 10\pi x - \pi/2)$$

Since x and t have same signs, the wave is travelling in negative x -direction.

b) the reference phase $\phi_0 = -\pi/2$

c) From eq of wave $y(x,t) = A \cos(\omega t + \beta x + \phi_0)$

$$\omega = 2\pi f$$

$$\omega = 4\pi$$

$$2\pi f = 4\pi$$

$$f = 2\text{Hz}$$

d) $\beta = 2\pi/\lambda$

$$10\pi = \frac{2\pi}{\lambda}$$

$$\lambda = 1/5 = \underline{0.2\text{m}}$$

e) phase velocity $= \omega/\beta = \frac{4\pi}{10\pi} = \frac{4}{10} = \frac{2}{5} = \underline{0.4\text{ m/s}}$

Problem 2

Give expressions for $y(x,t)$ for a sinusoidal wave travelling along a string in negative x -direction, given that $y_{\text{max}} = 40\text{cm}$, $\lambda = 30\text{cm}$, $f = 10\text{Hz}$ and

a) $y(x,0) = 0$ at $x = 0$

b) $y(x,0) = 0$ at $x = 3.75\text{cm}$

Ans - Wave equation for a sinusoidal wave is
 $y(x,t) = A \cos(\omega t + \beta x + \phi_0)$

$$A = y_{\max} = 40$$

$$\lambda = 30 \text{ cm}$$

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi \times 10^2}{30} = \frac{20\pi}{3}$$

$$f = 10 \text{ Hz}$$

$$\omega = 2\pi f = 20\pi$$

$$y(x,t) = 40 \cos\left(20\pi t + \frac{20\pi x}{3} + \phi_0\right)$$

a) $y(x,0) = 0, x = 0$

$$40 \cos(\phi_0) = 0$$

$$\phi_0 = \pm \pi/2$$

$$\left\{ \begin{array}{l} y(x,t) = -40 \sin\left(20\pi t + \frac{20\pi x}{3}\right) \text{ (cm)} \\ \phi_0 = +\pi/2 \end{array} \right\}$$
$$\left\{ \begin{array}{l} y(x,t) = 40 \sin\left(20\pi t + \frac{20\pi x}{3}\right) \text{ (cm)} \\ \phi_0 = -\pi/2 \end{array} \right\}$$

b) $\lambda = 3.75 \text{ cm} = 0.0375 \text{ m}$

$$y = 0 = 40 \cos\left(\frac{20\pi}{3} \times 0.0375 + \phi_0\right) = 0$$

$$\cos\left(\frac{\pi}{4} + \phi_0\right) = 0$$

$$\frac{\pi}{4} + \phi_0 = \pm \frac{\pi}{2}$$

$$\phi_0 = \frac{\pi}{4} \text{ or } -\frac{3\pi}{4}$$

$$y(x, t) = \begin{cases} 40 \cos(20\pi t + 20\pi/3 x + \pi/4) \text{ (cm)} & \phi_0 = \pi/4 \\ 40 \cos(20\pi t + 20\pi/3 x - 3\pi/4) \text{ (cm)} & \phi_0 = -3\pi/4 \end{cases}$$

Problem 3

The voltage of an electromagnetic wave traveling on a transmission line is given by

$$v(z, t) = 5e^{-\alpha z} \sin(4\pi \times 10^9 t - 20\pi z) \text{ (V)}, \text{ where}$$

z is the distance in metres from generator

a) Find the frequency, wavelength and phase velocity of the wave

b) At $z = 2 \text{ m}$, the amplitude of the wave was measured to be 2 V . Find α .

a) frequency

$$\omega = 2\pi f \text{ from eq}$$

$$\omega = 4\pi \times 10^9$$

$$\frac{2\pi f}{2} = 4\pi \times 10^9$$

$$\underline{\underline{f = 2 \text{ GHz}}}$$

wavelength

$$\beta = 20\pi$$

$$\frac{2\pi}{\lambda} = 20\pi$$

$$\underline{\underline{\lambda = 0.1 \text{ m}}}$$

$$\text{phase velocity} = \frac{\omega}{\beta} = \frac{4\pi \times 10^9}{20\pi} = \underline{\underline{2 \times 10^8 \text{ m/s}}}$$

b) $z = 5e^{-\alpha \times 2}$ — [Amplitude at $z = 2 \text{ m}$]

$$0.4 = e^{-2\alpha}$$

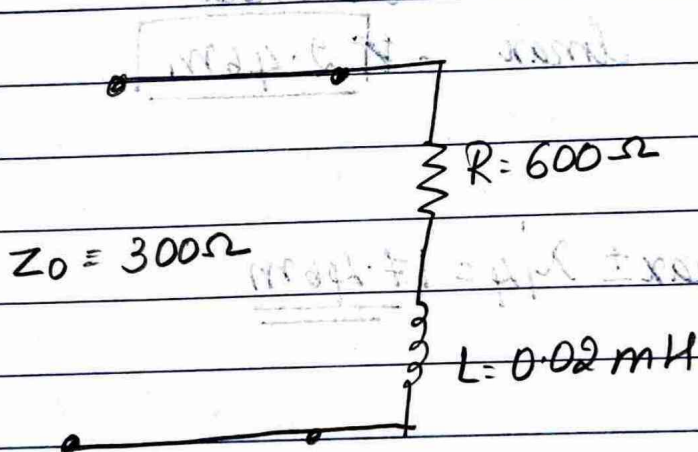
$$-0.9162 = -2\alpha$$

$$\underline{\underline{\alpha = 0.458}}$$

Problem 5

A 300Ω lossless transmission line is connected to a complex load composed of a resistor in series with an inductor as shown below. At 5 MHz determine a) reflection coefficient Γ

- VSWR
- location of voltage maxima nearest to load
- location of voltage minimum nearest to load



$$\begin{aligned}
 \text{a) } \Gamma &= \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{600 + 200j - 300}{600 + 200j + 300} \quad \left| \begin{array}{l} Z_L = 600 + j\omega L \\ = 600 + j \times 0.02 \times 5 \times 2\pi \end{array} \right. \\
 &= \frac{300 + 200j}{900 + 200j} = \frac{695.97 \angle 64.46^\circ}{1097.44 \angle 34.90^\circ} = (600 + 200j)\Omega
 \end{aligned}$$

$$\boxed{0.634 \angle 29.56^\circ}$$

$$b) VSWR = \frac{1+|\Gamma|}{1-|\Gamma|} = \frac{1+0.634}{1-0.634} = \underline{\underline{4.46}}$$

$$c) \frac{\beta L - \theta}{2} = n\pi, \quad V \text{ is max}$$

$$n=0$$

$$L = \frac{0\pi}{2\beta} = \frac{0\lambda}{4\pi} =$$

$$\theta = 29.56^\circ = 0.516 \text{ rad}$$

$$l_{\max} = \underline{\underline{2.46 \text{ m}}}$$

$$d) l_{\min} = l_{\max} \pm \lambda/4 = \underline{\underline{1.746 \text{ m}}}$$

Problem 6

At an operating frequency of 300 MHz, a lossless 50Ω air spaced transmission line 2.5 m in length is terminated with an impedance $Z_L = 40 + 20j\Omega$. Find a) input impedance

b) the fraction of input power delivered to the load

Ans: a) $Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)} \right]$

$$\beta l = \frac{2\pi \times 2.5}{\lambda} = \frac{2\pi \times 2.5}{10^2 \times 3 \times 10^6 / 3 \times 10^8} = 5\pi$$

$$\tan(\beta l) = 0$$

$$Z_{in} = \frac{Z_0 \times Z_L}{Z_0} = Z_L = 40 + 20j\Omega$$

b) $P_{del} = P_{in}(1 - |\Gamma|^2)$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{-10 + 20j}{90 + 20j} = 0.243 \angle 50.9^\circ$$

$$\frac{P_{del}}{P_{in}} = 1 - 0.243^2 = 0.941 = \underline{94.1\%}$$

Problem 7

A $75\text{-}\Omega$ lossless line is terminated in a load impedance $Z_L = R_L + jX_L$.

a) What must be the relationship between R_L and X_L in order that the standing wave ratio on the line is 3?

b) Find X_L , if $R_L = 150\text{-}\Omega$

c) Where does the voltage minimum nearest to the load occur for part b?

$$a) \Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{R_L - 75 + jX_L}{R_L + jX_L + 75}$$

$$S = |\Gamma| = \frac{\sqrt{(R_L - 75)^2 + X_L^2}}{\sqrt{(R_L + 75)^2 + X_L^2}}$$

$$\text{vs } WR = \frac{1+S}{1-S} = 3 \rightarrow S = \frac{S-1}{S+1} = 0.5$$

$$S^2 = \frac{1}{4} = \frac{(R_L - 75)^2 + X_L^2}{(R_L + 75)^2 + X_L^2}$$

$$R_L^2 + 75^2 + 2 \cdot 150R_L + X_L^2 = 4R_L^2 + 22500 - 600R_L + 4X_L^2$$
$$-3R_L^2 - 16875 + 750R_L = 3X_L^2$$

$$X_L = \sqrt{-R_L^2 - 5625 + 250R_L} = X_L$$

b) if $R_L = 150 \Omega$, $X_L = \pm \underline{96.824 \Omega}$

c) if $X_L = +96.824 \Omega$

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{150 + j96.8 - 75}{150 + j96.8 + 75} = 0.5 \angle 29^\circ$$

$\theta = 29^\circ = 0.506 \text{ rad}$

$\theta - 2\beta l = -\pi$

$$\frac{\pi + \theta}{2\beta} = \frac{(\pi + \theta)\lambda}{4\pi} = \frac{(3.14 + 0.506) \times \lambda}{4\pi}$$

$$= \underline{0.290\lambda}$$

if $X_L = -96.824 \Omega$

$$\Gamma = \frac{75 - j96.8}{225 - j96.8} = \frac{122.45 \angle -52.23^\circ}{244.93 \angle -23.27^\circ} = 0.49 \angle -28.96^\circ = \underline{0.505 \text{ rad}}$$

$$l = \frac{(\pi + \theta)\lambda}{4\pi} = \underline{0.209\lambda}$$

Problem 8

Consider a transmission line with line impedance Z_0 terminated with an unknown load Z_L . Show that:

$$Z_L = Z_0 \frac{1 - jS \tan(\beta l_{\min})}{S - j \tan(\beta l_{\min})}$$

where S is the SWR and $l_{\min}$ is the length from the load to the first voltage minimum

Ans: $Z_{\min} = Z_0 / S$

$$Z_{\min} = Z_0 \left[\frac{Z_L + jZ_0 \tan(\beta l_{\min})}{Z_0 + jZ_L \tan(\beta l_{\min})} \right]$$

$$\frac{Z_0}{S} = Z_0 \left[\frac{Z_L + jZ_0 \tan(\beta l_{\min})}{Z_0 + jZ_L \tan(\beta l_{\min})} \right]$$

$$Z_0 + jZ_L \tan(\beta l_{\min}) = Z_L S + jZ_0 S \tan(\beta l_{\min})$$

$$Z_0 - jZ_0 S \tan(\beta l_{\min}) = Z_L S - jZ_L \tan(\beta l_{\min})$$

$$Z_0 - Z_L [S - j \tan(\beta l_{\min})]$$

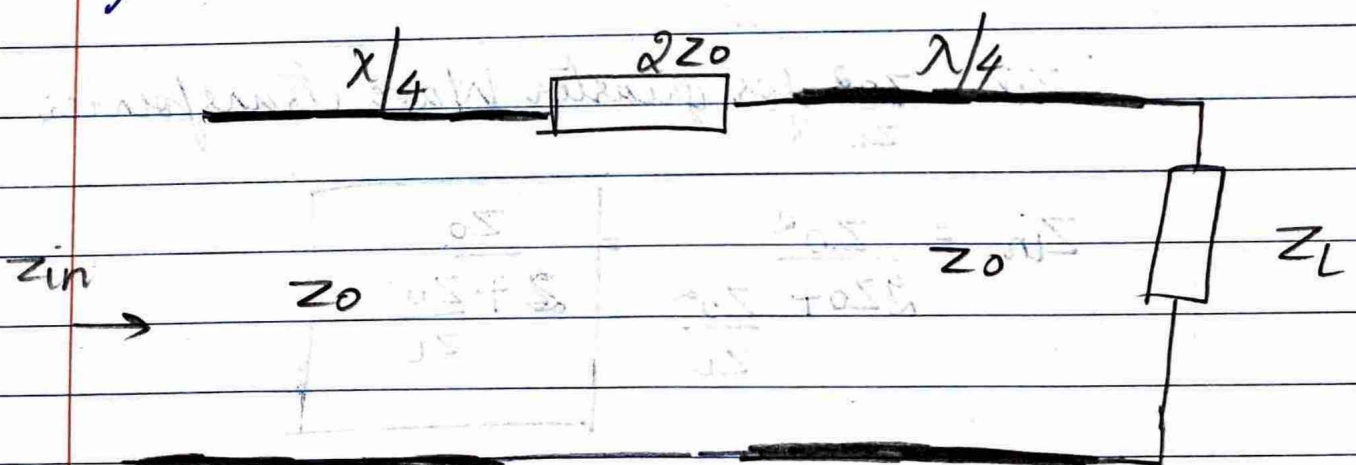
$$Z_L = \frac{Z_0 - jZ_0 S \tan(\beta l \sin)}{S - j \tan(\beta l \sin)}$$

$$Z_L = Z_0 \left[\frac{1 - jS \tan(\beta l \sin)}{S - j \tan(\beta l \sin)} \right]$$

Hence proved

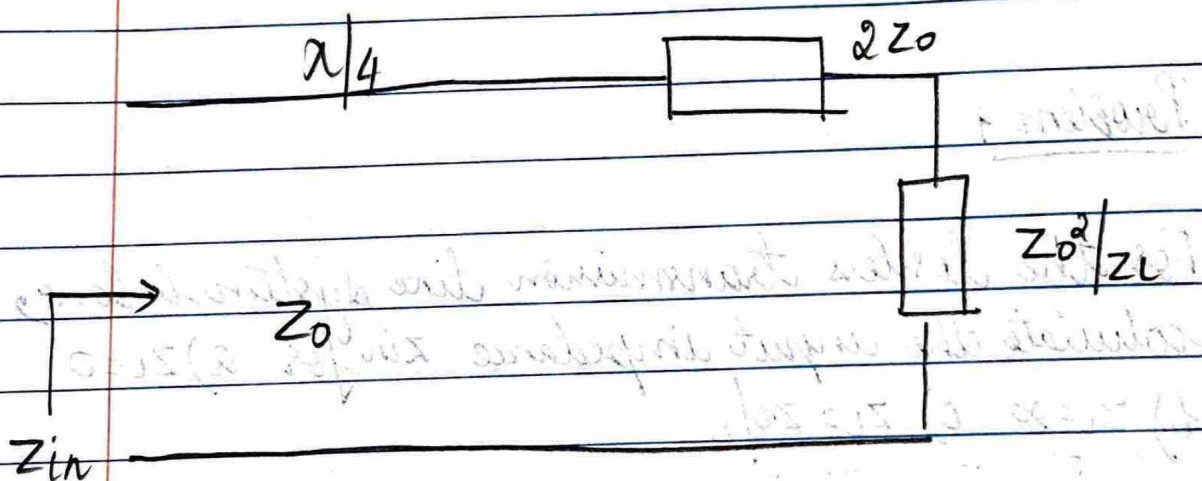
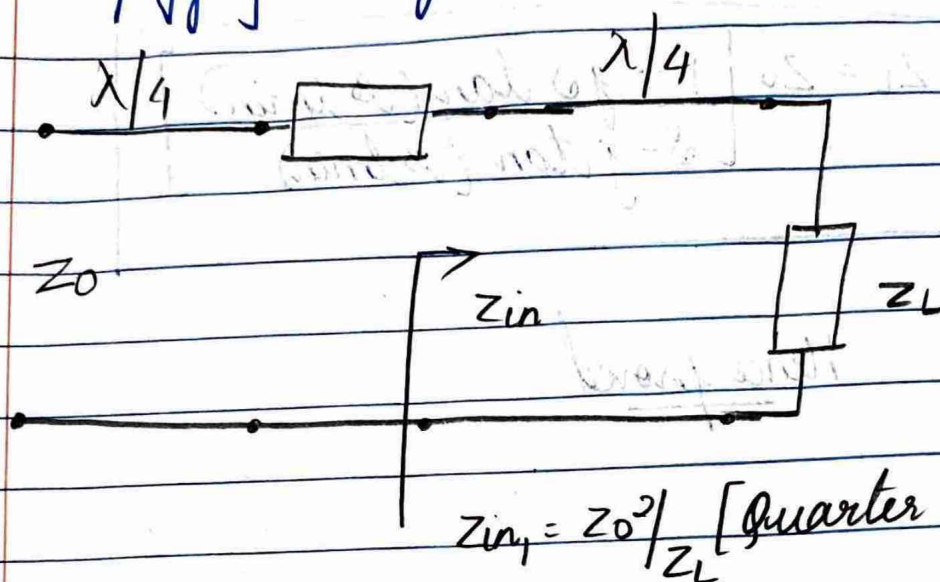
Problem 9

For the lossless transmission line system below, calculate the input impedance Z_{in} for a) $Z_L = 0$
b) $Z_L = \infty$ c) $Z_L = Z_0/2$



Ans. -

simplifying the given network



$Z_{in} = \frac{Z_0^2}{Z_L}$ for Quarter Wave transformer

$$Z_{in} = \frac{Z_0^2}{2Z_0 + \frac{Z_0^2}{Z_L}} = \boxed{\frac{Z_0}{2 + \frac{Z_0}{Z_L}}}$$

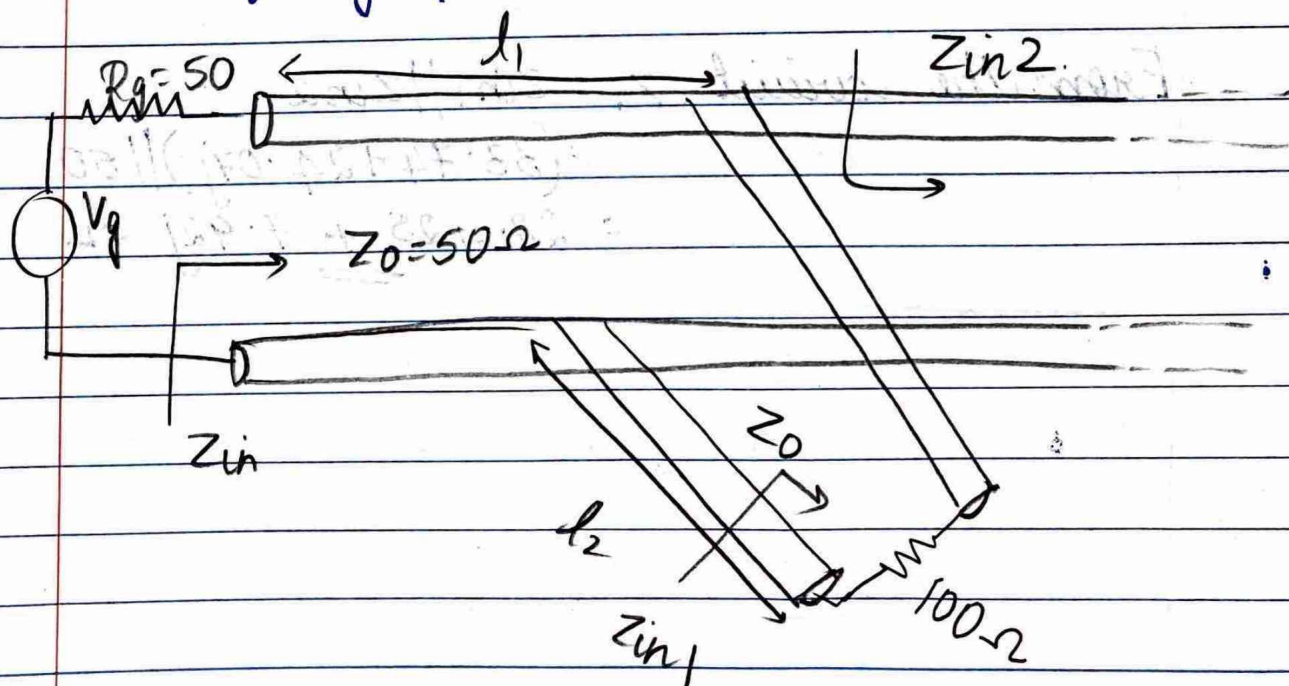
$$a) Z_L = 0 \Omega \quad Z_{in} = \frac{Z_0}{2 + \frac{Z_0}{0}} = \frac{Z_0}{\infty} = \underline{\underline{0 \Omega}}$$

$$b) Z_L = \infty \quad Z_{in} = \frac{Z_0}{2 + 0} = \underline{\underline{\frac{Z_0}{2} \Omega}}$$

$$c) Z_L = \frac{Z_0}{2} \quad Z_{in} = \frac{Z_0}{2 + \frac{Z_0 \times 2}{Z_0}} = \underline{\underline{\frac{Z_0}{4} \Omega}}$$

Problem 10

Consider the transmission line system shown below operating $f = 1 \text{ GHz}$. The wave travels at $2 \times 10^8 \text{ m/s}$. ($l_2 = 17 \text{ cm}$, $l_1 = 25 \text{ cm}$). Find input impedance at the beginning of transmission line (Z_{in})



Ans:- $\lambda = v/f = \frac{2 \times 10^8}{1 \times 10^9} = 0.2 \text{ m} = 20 \text{ cm}$

$l_1 = 25 \text{ cm} = \frac{25}{20} \lambda = \frac{5}{4} \lambda$

$l_2 = 17 \text{ cm} = \frac{17}{20} \lambda = 0.85 \lambda$

$Z_{in2} = Z_0 = 50 \Omega$ (infinite line)

$Z_{in1} = Z_0 \left[\frac{Z_L + Z_0 j \tan(\beta l)}{Z_0 + Z_L j \tan(\beta l)} \right]$ for input impedance

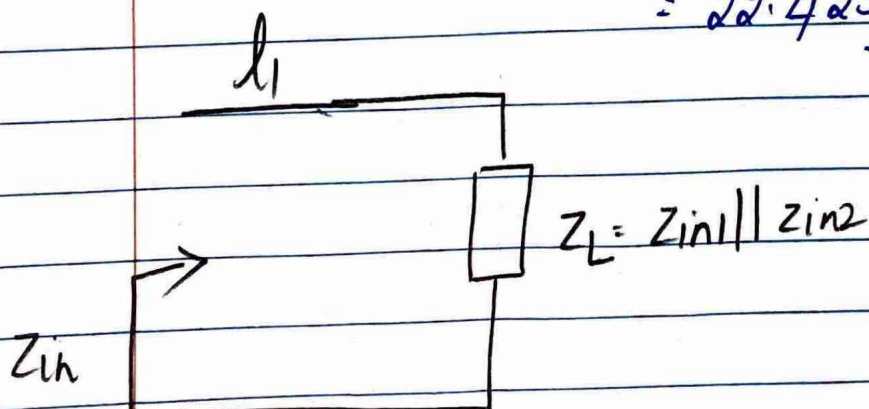
$= 50 \left[\frac{100 + j50 \tan(2\pi/\lambda \times 0.85\lambda)}{50 + 100j \tan(2\pi/\lambda \times 0.85\lambda)} \right]$

$Z_{in1} = \underline{\underline{33.74 + j24.07 \Omega}}$

From the circuit $Z_L = Z_{in1} \parallel Z_{in2}$

$= (33.74 + j24.07) \parallel 50$

$= \underline{\underline{22.425 + j7.92 \Omega}}$



It is quarter wave transformer

$$5\lambda/4 = \lambda + \lambda/4$$

$$Z_{in} = \frac{Z_0^2}{Z_L} = \frac{50 \times 50}{22.425 + 7.92j} = \underline{(99.1 - 35j) \Omega}$$

Problem 11

A lossless quarter wave transmission line with Z_0 is terminated with a load $R_L + jX_L$. Prove that input impedance is effectively a resistance in parallel with a capacitance.

For a quarter wave transformer

$$Z_{in} = \frac{Z_0^2}{Z_L}$$

$$Z_0^2 = Z_{in} Z_L = Z_{in} (R_L + jX_L)$$
$$Z_0 = \sqrt{Z_{in} (R_L + jX_L)}$$
$$Z_{in} = \frac{Z_0^2}{R_L + jX_L}$$

$$\frac{Z_0^2 (R_L - jX_L)}{R_L^2 + X_L^2} = \frac{Z_0^2 R_L}{R_L^2 + X_L^2} - j \frac{Z_0^2 X_L}{R_L^2 + X_L^2}$$

The imaginary part represents a capacitance as it is negative.

when $R \parallel C$ $Z = \frac{1}{\frac{1}{R} + j\omega C}$ $X_{in} = -\frac{1}{\omega C}$
 Equating the imaginary parts $C = \frac{1}{\omega |X_{in}|}$

$$C = \frac{R L^2 + X_L^2}{\omega Z_0^2 X_L}$$

Problem 12

given $L = 3 \text{ H/m}$, $C = 20 \text{ F/m}$, $\omega = 5 \times 10^9 \text{ rad/s}$ for a transmission line, what is the lossless attenuation constant and propagation constant?

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

for lossless $R = 0$, $G = 0$

$$\gamma = \sqrt{(j\omega L)(j\omega C)}$$

$$= \sqrt{-\omega^2 LC}$$

$$\alpha = 0$$

$$\beta = \omega \sqrt{LC}$$

$$\gamma = \underline{j\omega \sqrt{LC}}$$

$$\beta = 5 \times 10^9 \sqrt{3 \times 20}$$

$$= 38.729 \times 10^9 \text{ rad/m}$$

$$\gamma = \alpha + j\beta$$

$$= 0 + \underline{38.729 \times 10^9 j}$$